

Local Multi-Agent Bioinformatics Analysis Agent for Vision (Ocular Biology) Research

Summer Internship Project Proposal – University of California, Irvine

1. Abstract

Current cloud-based AI platforms like Kosmos/Biomni require uploading large-scale and sensitive biomedical data for every analysis, lack reliable GPU access, and lose workflow history after extended use. For ocular biology research — which relies on large single-cell RNA-seq, spatial transcriptomics, and multi-omics datasets from retina, RPE, and cornea — this creates cost, privacy, and reproducibility barriers.

This project will develop a local, chatbot-style bioinformatics analysis agent that executes computational workflows directly on institutional infrastructure. The system integrates a locally deployed open-source LLM (Qwen 3.6) with a multi-agent framework to coordinate analysis tasks, tool execution, and result generation without data leaving UCI.

A core innovation is persistent workflow memory: every analysis is versioned, reusable, and iteratively refinable, enabling both routine QC and advanced statistical modeling through a unified natural-language interface.

2. Brief Description of the Project

This project aims to develop a local, multi-agent bioinformatics analysis agent for research in vision (ocular biology). The system will provide a chatbot-style interface that allows researchers to execute computational workflows directly on institutional computing infrastructure without transferring sensitive biomedical data to external platforms.

The platform will support common bioinformatics workflows used in ocular biology, including:

- single-cell RNA-seq
- spatial transcriptomics
- multi-omics integration
- quality control
- differential expression analysis
- pathway enrichment
- statistical modeling

It will integrate locally deployed open-source large language models (e.g., Qwen 3.6) with a multi-agent framework that coordinates analysis tasks, tool execution, and result generation. A key feature is persistent workflow memory, enabling reproducibility, iterative refinement, and reuse of previous analyses. The system is designed to make routine and advanced bioinformatics analyses more accessible through a unified interface while efficiently utilizing local CPU and GPU resources.

3. System Infrastructure

Tier 1 – Lab Server (Coordinator):

- Ubuntu Server OS
- 36-Core CPU
- 256GB RAM
- 7TB Hard Drive

Good for: web UI, LLM coordinator, vector memory, skill library.

Tier 2 – UCI HPC3 / RCIC (Compute):

- NVIDIA A30, L40S, A100, RTX6000 GPU devices. Good for computing and hosting local LLM
- Hundreds of CPU computing nodes, good for computing

All analysis runs in-place on DFS/CRSP storage via Slurm – no data uploads required.

4. Required Skills / Programming Languages

- Familiarity with machine learning / LLM frameworks (e.g., PyTorch, Hugging Face)
- Experience with workflow orchestration or multi-agent systems, e.g., LangGraph / CrewAI
- Basic web development (e.g., FastAPI, React, or similar frameworks)
- Docker / containerization experience is required
- Preferred: Python/R bioinformatics, Slurm, Linux server administration

5. Potential Challenges & Contingency Plans

- Complexity of multi-agent coordination
Multi-agent orchestration introduces system complexity. We will start with a minimal agent framework and gradually extend to specialized modular agents for different analysis tasks.
- Reliability of open-source LLM outputs
Open-source models may produce variable outputs. We will mitigate this through structured prompting, tool-augmented execution, and validation of analytical results.

6. Expected Deliverables

- Working local chatbot deployed on lab server + HPC3
- Reusable ocular bioinformatics analysis skills
- Persistent workflow memory system
- Documentation and user guide